Chinese-to-English Technical Question Retrieval: Translation, Terminology, and Query Robustness

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1 Introduction (10 points)

Most high-quality technical Q&A, such as Stack Exchange, is in English, yet many developers describe problems more easily in their native language. For Chinese-speaking developers, translating technical terms into English is the main obstacle to searching it (Xu et al., 2018).

We study **cross-lingual technical question retrieval**: given a Chinese question, possibly containing English terms or commands, retrieve English posts that ask the same problem. Recent work favors multilingual dense retrieval over translation (Zuo et al., 2025; Goworek et al., 2025), but evaluates on general-domain text, and mixed-language queries are known to hurt retrievers (Zeng et al., 2026). We compare translation-based and direct retrieval on CQADupStack, test whether keeping terms in English helps, and analyze which queries fail.

This matters to non-English-speaking developers and to builders of retrieval-augmented assistants, whose answers depend on finding the right source. Unlike direct LLM answers, retrieved posts link to sources that users can check.

2 IR Task Definition (20 points)

This project aims to retrieve English community questions that describe the same technical problem as a Chinese query, comparing direct cross-lingual retrieval with query translation and examining how terminology preservation affects retrieval accuracy.

The main target users for this retrieval system are Chinese-speaking users searching for existing technical discussions written in English. These users may find it easier to describe the problems in Chinese, while retaining technical terms, error messages, commands etc, in English. Their information need is to locate an existing question that describes the same underlying problem, allowing them to examine its discussion and potential solutions.

This tasks requires more than matching a general topic. Queries and documents may share the same technical terminology while differing in details such as environment, operations, or essential constraints. A searching result should therefore return relevant results that match the specific problem rather than only discuss related concepts.

2.1 Inputs and Outputs

The system will take a Chinese technical question as input. A query may contain English terminologies and literal technical strings such as commands, error messages, and software names. The document collection consists of English questions from one selected technical subforum in CQADupStack, represented by their titles and bodies. The system returns a ranked list of the top-k relevant questions. Relevance is defined as describing the same underlying problem, including its essential conditions and requested operation, and is evaluated primarily using the dataset's community duplicate-question annotations.

2.2 Example Queries and Relevance

The following illustrative example shows how the same information need can be expressed in different languages:

A potentially relevant document would explain how to increase the memory allocated to WSL 2. In contrast, a document about reducing an application' s memory usage may address a similar symptom but a different goal.

Query version	Example
EN-orig	How can I increase the memory allocated to WSL 2?
ZH-full	如何增加分配给 WSL 2 的内存，以解决内存不足的问题?
ZH-mixed	如何增加分配给 WSL 2 的 memory，以解决 memory 不足的问题?

Table 1: Example queries.

Relevance also depends on the environment: guidance for WSL 1 may not apply to WSL 2. This example illustrates the need to match Chinese and English terminology while preserving the requested operation and software environment.

2.3 Research Questions

This project investigates three research questions:

1. How does direct Chinese-to-English dense retrieval compare with translating Chinese queries into English before retrieval, using the same dense retriever?

2. How do retaining English technical terms and applying terminology-preserving translation affect lexical and dense retrieval performance?

3. Which query characteristics are associated with retrieval failures, and what evidence distinguishes translation errors from retrieval errors?

To support controlled comparisons, each selected information need will have an original English query, a Chinese version using appropriate Chinese terminology, and a Chinese version retaining corresponding English terminology. These versions should preserve the same meaning and essential conditions. The original English query serves as a reference condition for examining performance differences.

3 Data (20 points)

3.1 Data Sources

Our primary collection is **CQADupStack** (Hoogeveen et al., 2015) from the BEIR benchmark (Thakur et al., 2021), loaded via `ir_datasets`. Documents are English Stack Exchange question posts (title + body), and queries are the titles of other questions. A document is relevant if the community marked it as a *duplicate* of the query, i.e., it asks the same technical problem. We use three subforums: *android*, *programmers*, and *unix*. Every query has at least one relevant document, and all judged documents are present in the corpus.

Since no Chinese queries exist, we translate the English queries and keep their IDs to reuse the labels. Duplicate labels are incomplete, so we will also annotate the pooled top-10 results of 30 test queries on a 0/1/2 scale (same problem / helpful / not relevant), with two annotators per pair.

As a secondary, out-of-domain set, we pair the Chinese queries of **mMARCO** (Bonifacio et al., 2021) with the original English MS MARCO passages, which share query IDs and labels. Backups are the Stack Exchange data dump (duplicate links in `PostLinks`) and LoTTE (Santhanam et al., 2022).

3.2 Data Pre-processing

For documents, we strip HTML, keep code and error messages verbatim, and index title + body. Each English query is translated by an LLM into two Chinese versions: **ZH-full** (technical terms translated) and **ZH-mixed** (technical terms kept in English). In both, commands, identifiers, error messages, and version numbers stay unchanged. We manually check about 150 translations for changed goals, dropped constraints, or altered identifiers, and fix any errors. We also write 30–50 natural Chinese rewrites without seeing the relevant documents.

We tag queries with regular expressions for commands/flags, API identifiers, error messages, and version/OS mentions, and spot-check the tags by hand. Since BEIR provides only a test set, we split queries 20/80 into dev and test, stratified by subforum (about 529 dev and 2,118 test queries). For mMARCO, we sample 500–1,000 dev queries and re-rank BM25 top-1000 candidates to keep dense encoding affordable.

3.3 Data Examples

Table 2 shows two real queries with their planned Chinese versions and labeled duplicates. Labels can be loose: in *android*, *List of ROMs for HTC Flyer* is marked as a duplicate

of *(how) can I install android on an iphone 4?*, which motivates our supplementary annotation. In mMARCO, a query pair looks like 每天走多少步健康 / *how many steps a day healthy.*

unix (Q 111650)	
EN-orig	How to set up wireshark with correct permissions
ZH-full	如何为 wireshark 设置正确的权限?
ZH-mixed	如何为 wireshark 设置正确的 permissions?
Relevant	Can't get Wireshark to run as user on CentOS
programmers (Q 61558)	
EN-orig	Turn away a bug if no reproducible test case exists?
ZH-full	没有可复现的测试用例时，是否可以拒绝处理该缺陷?
ZH-mixed	没有可复现的 test case 时，是否可以拒绝处理该 bug?
Relevant	How do I isolate difficult to reproduce bugs?

Table 2: Example queries and labeled duplicates from CQADupStack.

3.4 Dataset Statistics

Table 3 summarizes the three subforums. Queries are short (median 8–9 words), while documents are much longer (median 83–140 words). Most queries have exactly one duplicate, but about a quarter have two or more.

Subforum	Docs	Queries	Qrels	Rel/Q	≥2 rel	Q len	D len
android	22,998	699	1,696	2.43	27.3%	9	83
programmers	32,176	876	1,675	1.91	28.3%	9	140
unix	47,382	1,072	1,693	1.58	25.3%	8	101
Total	102,556	2,647	5,064	1.91	26.8%	–	–

Table 3: CQADupStack statistics. Rel/Q: mean relevant documents per query; ≥2 rel: share of queries with two or more; lengths are median words.

Table 4 shows that explicit technical strings are rare. Commands and flags appear almost only in *unix* (2.8%), and no feature exceeds 8% in any subforum. Most queries are therefore natural-language titles, so the ZH-full vs. ZH-mixed contrast applies to many queries, while identifier protection affects a small subset. We will report per-feature results as exploratory because these groups are small.

For mMARCO, the dev/small set has 6,980 Chinese queries over 8.8M English passages, with about one relevant passage per query.

Subforum	Cmd/flag	API id.	Error	Ver./OS
android	0.3%	7.7%	4.7%	6.0%
programmers	0.0%	2.1%	2.1%	0.6%
unix	2.8%	3.6%	4.1%	5.7%

Table 4: Share of queries with each feature (regex-based, approximate).

4 Related Work (10 points)

Translation vs. multilingual retrieval: Two recent studies compare the main routes for cross-lingual retrieval (CLIR). Zuo et al. (Zuo et al., 2025) evaluate LLM rerankers in two-stage CLIR and find that multilingual bi-encoders are better first-stage retrievers than BM25 over machine-translated text, and that the benefit of translation shrinks as rerankers get stronger; without translation, however, current rerankers fall well short. Goworek et al. (Goworek et al., 2025) compare document translation, multilingual dense retrieval, contrastive alignment, and cross-encoder reranking on three benchmarks, and find that dense models trained for CLIR beat lexical matching and gain little from translation. Both use general-domain benchmarks (news, Wikipedia, web passages). We test whether these conclusions hold for developer questions, where an exact command or API name can decide relevance and translation may help lexical matching more than these studies suggest. We also hold the retriever fixed when comparing translated and untranslated queries, which separates the effect of translation from that of the retriever.

Multilingual embedding models: Chen et al. (Chen et al., 2024) present BGE-M3, which produces dense, sparse, and multi-vector representations from one model for over 100 languages and achieves state-of-the-art results on multilingual and cross-lingual benchmarks. We use it off the shelf as our direct cross-lingual retriever; its built-in sparse mode also lets us test lexical and semantic matching within one model.

Code-switched queries: Zeng et al. (Zeng et al., 2026) build CSR-L, a human-annotated benchmark of naturally code-switched queries including English–Chinese, and CS-MTEB, which covers 11 tasks. Code-switching de-

graded statistical, dense, and late-interaction retrievers alike, with drops of up to 27%, which they trace to embedding divergence between pure and mixed text. Their queries come from general benchmarks and are mixed for naturalness; we study a specific, common form of mixing in technical questions, where only domain terms stay in English, and test whether it helps or hurts compared with fully Chinese queries for each retriever type.

LLM-based query expansion: Macmillan-Scott et al. (Macmillan-Scott et al., 2025) evaluate multilingual LLMs that expand cross-lingual queries by generating pseudo-documents. They find the largest gains for languages with the weakest baselines, while retrieval across different scripts remains especially poor. This motivates one of our candidate improvements: having an LLM write a hypothetical English post for a Chinese question before retrieval.

Technical-domain CLIR: Early work by Xu et al. (Xu et al., 2018) addressed our exact setting, retrieving English Stack Overflow questions for Chinese questions with domain-specific term translation, and identified term translation as the main bottleneck; it predates neural retrievers and was evaluated with a small user study. More recently, the TREC 2023 NeuCLIR track (Lawrie et al., 2024) added a pilot task that retrieves Chinese technical abstracts with English topics. Its direction and documents (academic abstracts) differ from ours: community Q&A posts that mix prose with commands, API names, and error messages.

Our position: Recent CLIR studies favor multilingual dense retrieval, but evaluate on general-domain text, while the only work on developer Q&A predates neural methods. We bring the two together: Chinese-to-English retrieval over Stack Exchange with modern retrievers, controlled query versions that isolate the language of technical terms, and failure analysis by query feature. We expect hybrid retrieval to beat either route alone, because it combines exact matching of identifiers with cross-lingual semantic matching, and term-preserving translation to beat plain translation, because it avoids the term errors Xu et al. identified.

5 Evaluation and Results (30 points)

5.1 Metrics

A developer searching in Chinese wants a duplicate of their problem near the top, and a downstream RAG system wants the right post somewhere in the candidate list. We therefore use three standard metrics, computed with `ir_measures` on the duplicate labels:

- **nDCG@10** (primary): ranking quality of the first page, the standard metric for CQADupStack in BEIR (Thakur et al., 2021), which makes our English results comparable to published ones.
- **MRR@10**: how soon the first duplicate appears. Most queries have exactly one duplicate (Table 3), so this closely matches the user's experience.
- **Recall@100**: whether the duplicate is retrieved at all, which bounds what a reranker or RAG generator can use.

On the 30 queries with supplementary labels, we also report graded nDCG@10 (0/1/2) and the share of top-10 results judged at least helpful, to estimate how much the incomplete duplicate labels understate performance. We record average query latency and translation cost per query, since a translation step adds both.

Results are reported per subforum and macro-averaged. We test differences between systems with a paired two-sided randomization test over queries ($p < 0.05$, Holm correction for multiple comparisons).

5.2 Baselines

- **Random**: each query receives a random ranking of the subforum's documents, averaged over 5 seeds.
- **Translate + BM25**: the Chinese query is machine-translated into English and searched with Pyserini BM25 using default parameters ($k_1 = 0.9$, $b = 0.4$), with no tuning.

As reference conditions (not upper bounds), we also run BM25 and dense retrieval with the original English queries. Our English BM25 should reproduce the published BEIR score (nDCG@10 $\approx$ 0.30 on CQADupStack), which checks our pipeline.

5.3 Systems Compared

Table 5 lists our runs. Pairing each query version with each retriever lets us separate the effects of translation, the retriever, and term language (RQ1–RQ2).

Query input	Retriever	Role
–	Random	Baseline
ZH → MT	BM25 (default)	Baseline
EN-orig	BM25 / BGE-M3	Reference
ZH-full, ZH-mixed	BGE-M3 (no translation)	Direct CLIR
ZH → MT	BGE-M3	Translate + dense
ZH-full, ZH-mixed	BM25 + BGE-M3 (RRF)	Hybrid
Best of above	+ multilingual reranker	Reranking
Term-preserving MT / HyDE	Best pipeline	Our improvements

Table 5: Runs. MT: machine translation of the Chinese query into English. RRF: reciprocal rank fusion (Cormack et al., 2009).

Our Chinese queries were themselves translated from English, so translating them back may simply recover the original title and inflate the translation route. We use a different model for ZH→EN than for EN→ZH, report word overlap between back-translations and the original titles, and repeat the main comparison on the natural rewrites, which do not have this bias.

5.4 Analysis

For RQ3, we break results down by query feature (commands, identifiers, error messages, version/OS, translatable terms) and by subforum, treating small groups as exploratory. We will also sample 50 test queries where the best Chinese run fails but the English reference succeeds, and label each failure as a translation error, term mismatch, missed fine-grained distinction (e.g., detect vs. prevent), or label noise. All tuning (RRF weights, reranking depth, prompts) is done on the dev split, and the test split is evaluated once.

6 Work Plan (10 points)

6.1 Completed So Far

We have (1) defined the task and research questions; (2) reviewed related work on cross-lingual, code-switched, and technical-domain retrieval; (3) loaded CQADupStack (*android*, *programmers*, *unix*) and mMARCO with `ir_datasets`; and (4) computed the statistics in Section 3. That check confirmed every query has at least one judged duplicate and every judged document is in the corpus. It also showed that explicit technical strings are rare in queries, which led us to center RQ2 on translatable technical terms. We also read sampled query–duplicate pairs and found noisy labels, which motivates the supplementary annotation.

6.2 Remaining Work

W1 (Oct 13–19) Build BM25 indexes for the three subforums and run the random and English BM25 runs. Write the translation prompt and translate all 2,647 queries into ZH-full and ZH-mixed.

W2 (Oct 20–26) Manually check 150 translations and fix errors, create the 20/80 dev/test split, tag query features, and count posts containing the "Possible Duplicate" banner.

W3 (Oct 27–Nov 2) Write 30–50 natural Chinese rewrites. Back-translate the Chinese queries with a second model and run translate + BM25. Encode all documents with BGE-M3 into FAISS, and run direct dense retrieval (ZH-full, ZH-mixed) and the English dense reference.

W4 (Nov 3–9) Run translate + dense and hybrid RRF, tuning the fusion weight on dev, and write the progress check.

W5 (Nov 10–16) Add a multilingual reranker on the top-100 results of the best pipeline, implement improvement 1 (term-preserving translation), and set up significance testing.

W6 (Nov 17–23) Implement improvement 2 (HyDE) or an ablation, and annotate the pooled top-10 results of 30 test queries with two annotators.

W7 (Nov 24–30) This Thanksgiving week is a buffer for delays. If we are on schedule, we run the mMARCO extension (500 queries, re-ranking BM25 top-1000).

W8 (Dec 1–7) Freeze all settings and run the test set once. Break down results by query feature and subforum, and label 50 failure cases.

W9 (Dec 8–14) Draft the full report, including figures, tables, and discussion of RQ1–RQ3.

W10 (Dec 15–21) Revise the report, record the video, and release the code with a README.

References

Luiz Bonifacio, Vitor Jeronymo, Hugo Queiroz Abonizio, Israel Campiotti, Marzieh Fadaee, Roberto Lotufo, and Rodrigo Nogueira. 2021. mMARCO: A multilingual version of the MS MARCO passage ranking dataset. *arXiv preprint arXiv:2108.13897* .

Jianlyu Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*. pages 2318–2335.

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. 2009. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*. pages 758–759.

Roksana Goworek, Olivia Macmillan-Scott, and Eda B. Özyiğit. 2025. What drives cross-lingual ranking? retrieval approaches with multilingual language models. *arXiv preprint arXiv:2511.19324* .

Doris Hoogeveen, Karin M. Verspoor, and Timothy Baldwin. 2015. CQADupStack: A benchmark data set for community question-answering research. In *Proceedings of the 20th Australasian Document Computing Symposium (ADCS)*. pages 3:1–3:8. https://doi.org/10.1145/2838931.2838934.

Dawn Lawrie, Sean MacAvaney, James Mayfield, Paul McNamee, Douglas W. Oard, Luca Soldaini, and Eugene Yang. 2024. Overview of the TREC 2023 NeuCLIR track. *arXiv preprint arXiv:2404.08071* .

Olivia Macmillan-Scott, Roksana Goworek, and Eda B. Özyiğit. 2025. Generative query expansion with multilingual LLMs for cross-lingual information retrieval. *arXiv preprint arXiv:2511.19325* .

Keshav Santhanam, Omar Khattab, Jon Saad-Falcon, Christopher Potts, and Matei Zaharia. 2022. ColBERTv2: Effective and efficient retrieval via lightweight late interaction. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*. pages 3715–3734.

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*.

Bowen Xu, Zhenchang Xing, Xin Xia, David Lo, and Shanping Li. 2018. Domain-specific cross-language relevant question retrieval. *Empirical Software Engineering* 23(2):1084–1122. https://doi.org/10.1007/s10664-017-9568-3.

Qingcheng Zeng, Yuheng Lu, Zeqi Zhou, Heli Qi, Puxuan Yu, Fuheng Zhao, Hitomi Yanaka, Weihao Xuan, and Naoto Yokoya. 2026. Code-switching information retrieval: Benchmarks, analysis, and the limits of current retrievers. In *Findings of the Association for Computational Linguistics: ACL 2026*.

Longfei Zuo, Pingjun Hong, Oliver Kraus, Barbara Plank, and Robert Litschko. 2025. Evaluating large language models for cross-lingual retrieval. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. ArXiv:2509.14749.